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Module 4: Biodiversity, Evolution and Disease

OCR A A-Level Biology (H420)

4.1 Communicable Diseases

Types of Pathogens

A pathogen is a microorganism that causes disease in its host. Communicable (infectious) diseases are caused by pathogens that can be transmitted between organisms. The main types of pathogens are:

Pathogen Type	Characteristics	Examples
Bacteria	Prokaryotic cells; cause disease by producing toxins or destroying host tissue; reproduced by binary fission	Mycobacterium tuberculosis (TB), Vibrio cholerae (cholera), Helicobacter pylori (stomach ulcers)
Viruses	Non-living obligate intracellular parasites; contain DNA or RNA in a protein capsid; replicate using host cell machinery; destroy host cells on release	HIV (AIDS), Influenza, Tobacco mosaic virus (TMV)
Fungi	Eukaryotic organisms; many are saprophytes but some are parasitic; may produce spores for dispersal	Phytophthora infestans (potato blight), Candida (thrush), ringworm
Protoctists	Eukaryotic single-celled organisms; some are parasitic with complex life cycles involving vectors	Plasmodium (malaria, transmitted by Anopheles mosquito)

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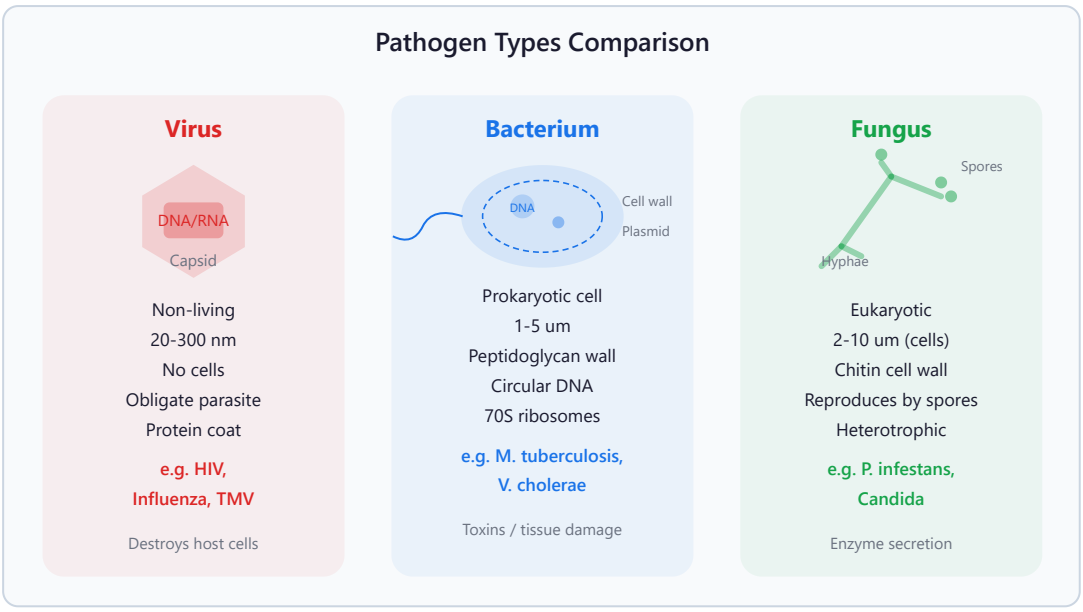

Modes of Transmission

Pathogens are transmitted between hosts through several routes:

- **Direct contact** — physical touching or exchange of body fluids (e.g., HIV through sexual contact or blood-to-blood, athlete's foot through skin contact)
- **Airborne (droplet infection)** — pathogens carried in tiny water droplets expelled by coughing or sneezing (e.g., TB, influenza)
- **Waterborne** — ingestion of contaminated water (e.g., cholera spreads through water contaminated with faeces)
- **Vector-borne** — transmitted by an organism (vector) that carries the pathogen without being affected (e.g., malaria via Anopheles mosquito bite)
- **Fomites** — inanimate objects contaminated with pathogens (e.g., door handles, surgical instruments)

Plant Diseases

Plants are susceptible to communicable diseases caused by bacteria, viruses, and fungi. Tobacco mosaic virus (TMV) causes mottled discolouration of leaves, reducing photosynthetic ability due to loss of chlorophyll. Ring rot (bacterial) causes wilting and rotting in potatoes. Black sigatoka (fungal) devastates banana crops by destroying leaf tissue.

Plants defend themselves through physical barriers (cellulose cell walls, waxy cuticle, bark), chemical defences (antimicrobial compounds, toxins such as cyanide in some species, tannins that inhibit

enzyme activity in herbivores), and the hypersensitive response (cells surrounding an infection site undergo rapid apoptosis, sealing off the pathogen and preventing its spread through the plant).

Disease Prevention and Control

Preventing communicable disease relies on breaking the chain of transmission. Methods include: improved sanitation and clean water supply (cholera), use of mosquito nets and insecticides (malaria), vaccination programmes (measles, TB), quarantine of infected individuals (TB), sterilisation of surgical equipment, barrier contraception (HIV), and public health education campaigns.

> **Exam Tip:** When discussing disease prevention, always link the method to the specific mode of transmission it interrupts. For example: "Mosquito nets prevent malaria transmission by creating a physical barrier between the human host and the Anopheles mosquito vector, preventing the bite through which Plasmodium sporozoites are introduced."

4.2 The Immune System

Non-Specific (Innate) Immunity

Non-specific defences act against all pathogens without distinguishing between them. They provide the first and second lines of defence:

First line of defence (barrier defences): skin (keratinised outer layer forms physical barrier; sebaceous glands produce antimicrobial oils), mucous membranes (mucus traps pathogens in respiratory and digestive tracts), stomach acid (pH 1-2 kills most ingested microorganisms), tears and saliva (contain lysozyme enzyme that digests bacterial cell walls), and blood clotting (seals wounds to prevent pathogen entry).

Second line of defence (cellular and chemical): phagocytosis (neutrophils and macrophages engulf and digest pathogens in phagosomes using lysosomal enzymes), inflammation (damaged cells release histamine causing vasodilation and increased capillary permeability, bringing more phagocytes to the site), fever (raised body temperature inhibits pathogen reproduction and enhances immune cell activity), and antimicrobial proteins (interferons inhibit viral replication in neighbouring cells; complement proteins lyse pathogen membranes).

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Specific (Adaptive) Immunity

The specific immune response targets individual pathogens based on their unique antigens (molecules on the pathogen surface that the immune system recognises as foreign). Adaptive immunity involves lymphocytes (B cells and T cells) and has two key properties: specificity (each lymphocyte responds to only one antigen) and immunological memory (faster, stronger response on re-exposure to the same pathogen).

> **Antigen:** A molecule (usually a protein or glycoprotein) on the surface of a cell that can be recognised by the immune system and trigger an immune response. Self-antigens (on the organism's own cells) are tolerated; non-self antigens (on pathogens or foreign cells) trigger an immune response.

Cell-Mediated Immunity (T Cells)

T lymphocytes mature in the thymus and respond to antigens presented on cell surfaces by antigen-presenting cells (APCs such as macrophages and dendritic cells). When a T cell receptor binds to a complementary antigen-MHC complex, the T cell is activated and clones rapidly by mitosis, differentiating into:

- **T helper cells (Th)** — release cytokines that activate B cells, stimulate phagocytosis, and coordinate the immune response
- **T killer cells (cytotoxic T cells, Tc)** — directly destroy infected host cells by releasing perforin (creating pores in cell membranes) and granzymes (triggering apoptosis)
- **T memory cells** — remain in the body long-term, enabling rapid secondary response on re-exposure
- **T regulatory cells (Treg)** — suppress the immune response after infection is cleared, preventing autoimmune damage

Humoral Immunity (B Cells)

B lymphocytes mature in the bone marrow and produce antibodies (immunoglobulins). When a B cell encounters its complementary antigen and receives cytokine signals from T helper cells, it is activated and undergoes clonal expansion (rapid mitosis), differentiating into:

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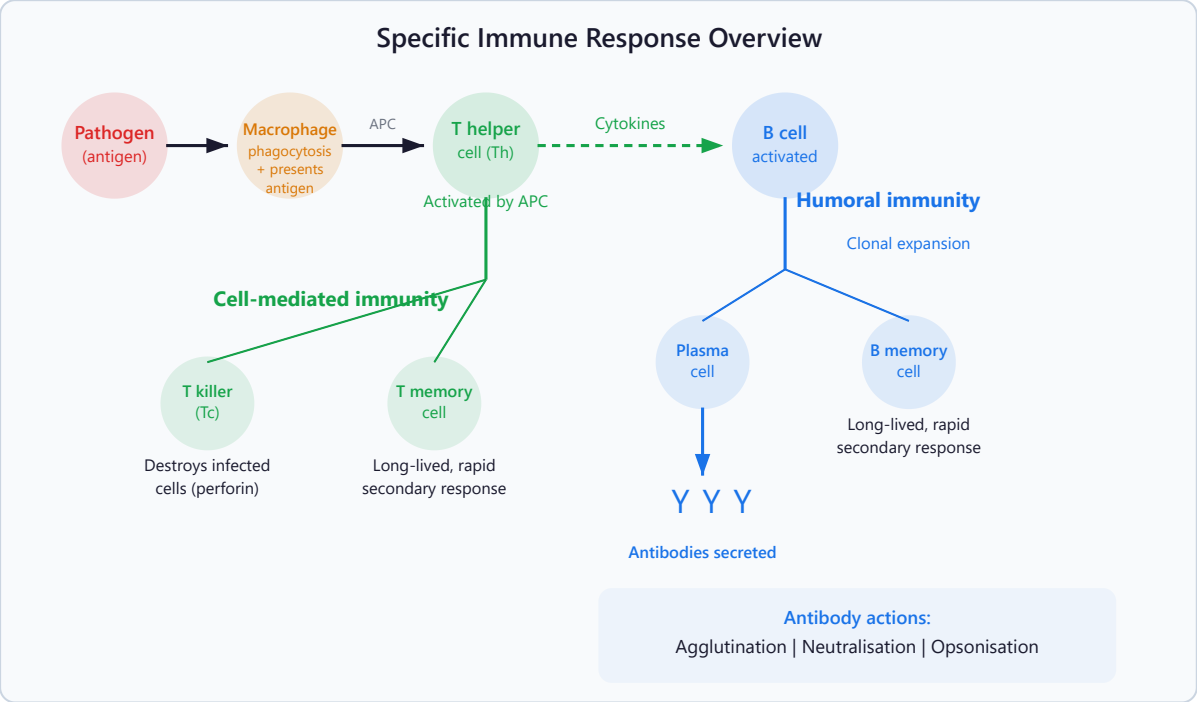
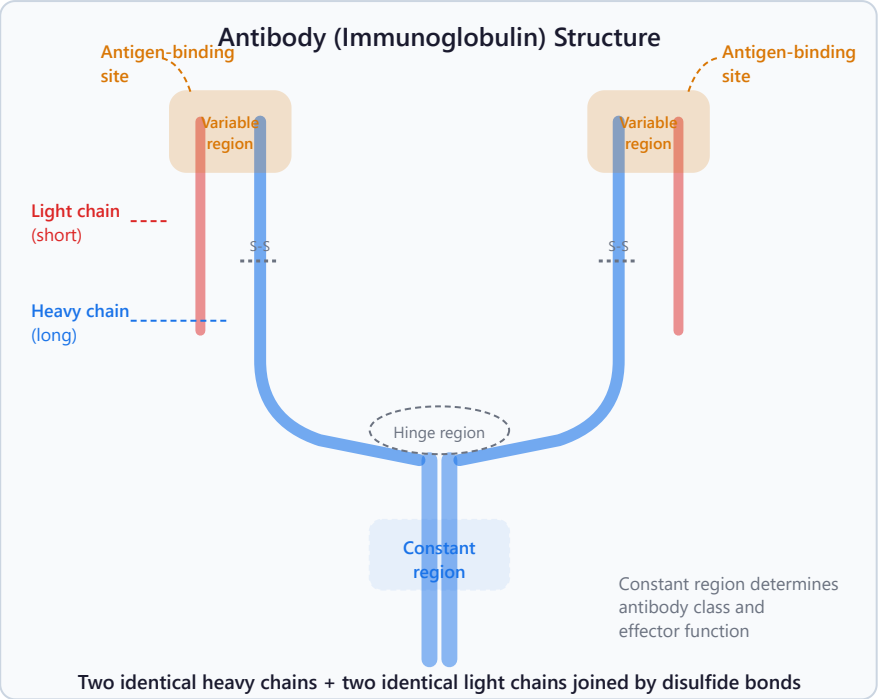
- **Plasma cells** — produce and secrete large quantities of specific antibodies (approximately 2000 molecules per second) into the blood and tissue fluid
- **B memory cells** — long-lived cells that remain in the body and mount a rapid, enhanced secondary immune response upon re-exposure to the same antigen



Antibody Structure and Function

Antibodies are Y-shaped glycoproteins consisting of two identical heavy chains and two identical light chains, joined by disulfide bonds. The variable regions at the tips of the Y form the antigen-binding sites, which are complementary in shape to a specific antigen (the shape is determined by the amino acid sequence of the variable region). The constant region determines the antibody class and its effector function.



Antibodies defend against pathogens by: agglutination (clumping pathogens together, preventing entry to cells and making them easier for phagocytes to engulf), neutralisation (blocking toxins or viral binding sites so they cannot interact with host cells), opsonisation (coating pathogens to enhance recognition and phagocytosis), and complement activation (triggering the complement cascade that lyses pathogen membranes).

Vaccination

Vaccination introduces a harmless form of an antigen (attenuated pathogen, dead pathogen, isolated antigen, or toxoid) to stimulate a primary immune response and the production of memory cells without causing disease. On subsequent exposure to the real pathogen, memory B and T cells mount a rapid, strong secondary immune response that eliminates the pathogen before symptoms develop.

Herd immunity occurs when a sufficiently high proportion of a population is vaccinated (typically 85-95% depending on the disease), reducing transmission to a level where even unvaccinated individuals are protected because the probability of encountering an infected person is very low. This protects vulnerable individuals who cannot be vaccinated (immunocompromised, very young, allergic to vaccine components).

HIV and the Immune System

Human immunodeficiency virus (HIV) is a retrovirus that infects and destroys T helper cells (CD4+ cells). The virus uses reverse transcriptase to convert its RNA genome into DNA, which is inserted into

the host cell genome. Over time, the progressive destruction of T helper cells severely compromises the immune system, leading to acquired immunodeficiency syndrome (AIDS), characterised by susceptibility to opportunistic infections and certain cancers that a healthy immune system would normally control.

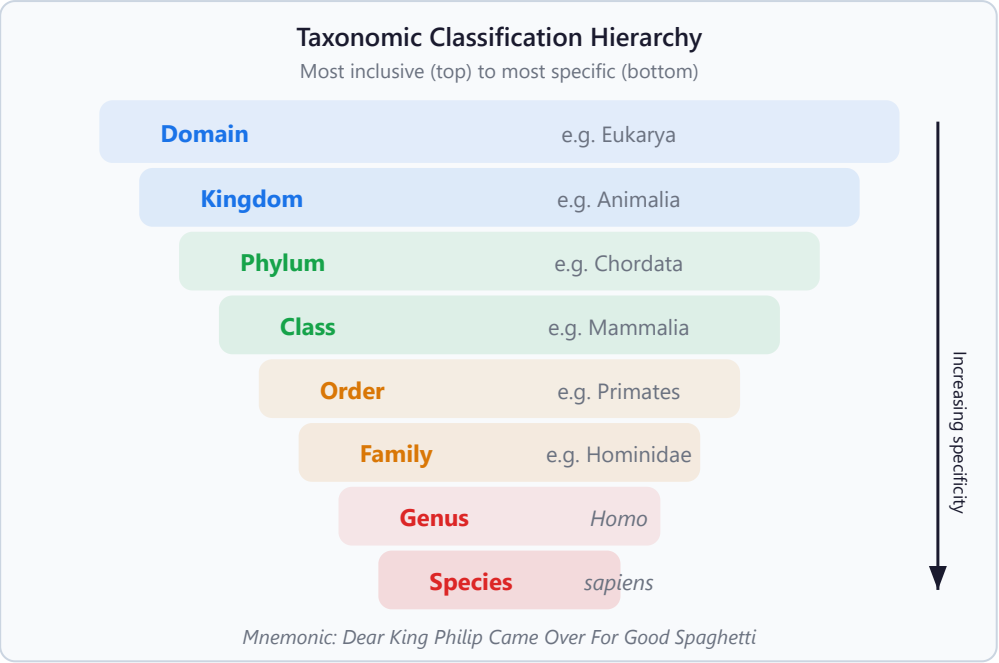
HIV is transmitted through exchange of body fluids (sexual contact, blood transfusion, sharing needles, mother-to-child during birth or breastfeeding). Prevention includes barrier contraception, screening blood products, needle exchange programmes, and antiretroviral therapy (ART) which suppresses viral replication and reduces transmission risk.

> **Exam Tip:** When describing the immune response, use the correct sequence: pathogen enters, non-specific response occurs first, antigen presentation activates specific response, clonal selection and expansion produce effector cells (plasma cells and T killer cells) and memory cells. Always distinguish between primary (slow, lower antibody titre) and secondary (fast, high antibody titre) responses.

4.3 Biodiversity

Classification

Taxonomy is the science of classifying organisms into hierarchical groups based on shared characteristics. The hierarchy from most inclusive to most specific is: Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species. The binomial naming system gives each species a two-part Latin name consisting of the genus (capitalised) and species (lowercase), both italicised.



The three-domain system (proposed by Carl Woese based on ribosomal RNA analysis) classifies all life into Bacteria, Archaea, and Eukarya. This replaced the five-kingdom system because molecular evidence revealed that Archaea are more closely related to Eukarya than to Bacteria, despite their superficial similarity as prokaryotes.

Phylogenetic classification groups organisms by evolutionary relationships rather than physical similarities. Molecular comparisons (DNA base sequences, amino acid sequences) and analysis of shared derived characteristics allow construction of phylogenetic trees (cladograms) showing evolutionary ancestry and divergence points.

Measuring Biodiversity

> **Biodiversity:** The variety of living organisms in an area, measured at three levels: genetic diversity (variation in alleles within a species), species diversity (number of different species and their relative abundance in a community), and ecosystem diversity (range of different habitats in an area).

Species richness is simply the number of different species present in a community. However, it does not account for the relative abundance of each species. **Simpson's Index of Diversity (D)** provides a more meaningful measure by incorporating both species richness and evenness:

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D = 1 - (sum of (n/N)^2)

Where:
n = number of individuals of each species
N = total number of all individuals
D ranges from 0 (no diversity) to 1 (infinite diversity)
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A higher value of D indicates greater biodiversity. Communities with high species richness and even distribution of individuals among species will have the highest diversity indices.

Sampling Methods

Sampling techniques allow estimation of population sizes and species distributions without counting every individual. Random sampling uses randomly generated coordinates to place quadrats, avoiding investigator bias. Systematic sampling places quadrats at regular intervals along a transect to study changes across an environmental gradient (zonation).

The **Lincoln index** estimates population size of mobile animals using mark-release-recapture: Population size = (number in first capture x number in second capture) / number marked in second capture. Assumptions include: no immigration/emigration, no births/deaths between captures, marks do not affect survival or behaviour, and sufficient time for mixing.

4.4 Evolution and Natural Selection

Natural Selection

Darwin's theory of evolution by natural selection explains how populations change over time in response to environmental pressures. The mechanism requires: genetic variation within a population (caused by mutation, meiosis, and sexual reproduction), overproduction of offspring (more offspring produced than can survive), competition for limited resources (food, mates, territory), differential survival and reproduction (individuals with advantageous phenotypes are more likely to survive and reproduce), and inheritance (advantageous alleles are passed to the next generation, increasing their frequency in the population over many generations).

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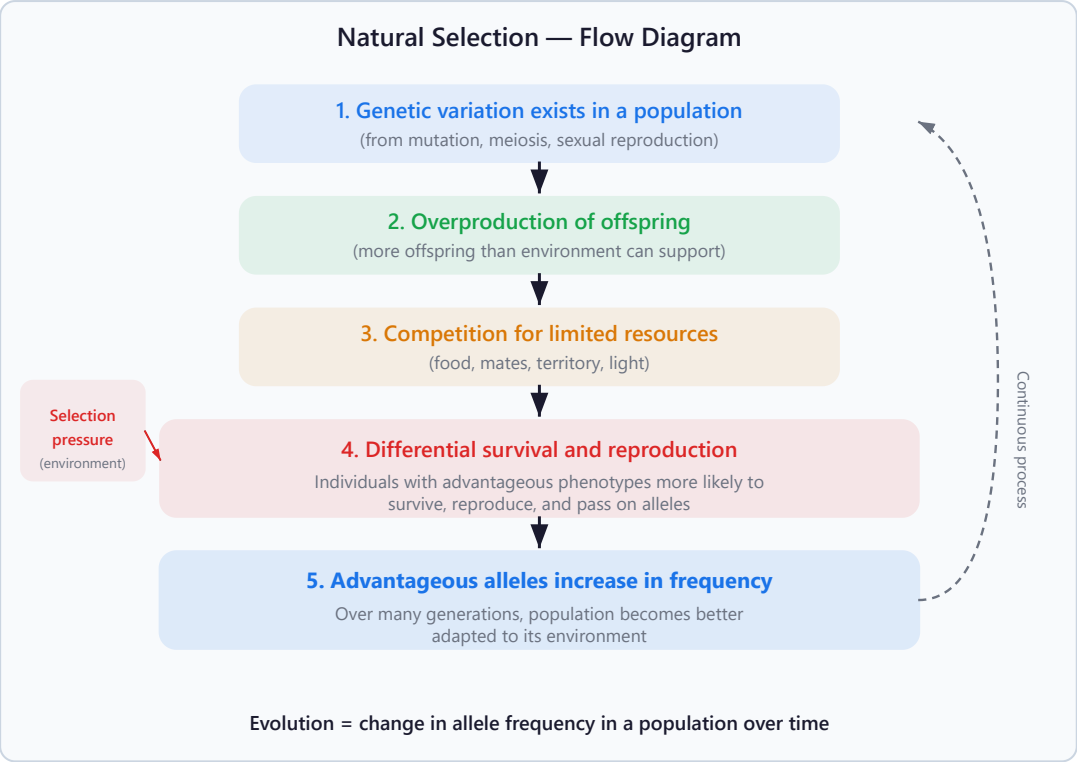

Types of Selection

- **Stabilising selection** — individuals with phenotypes close to the mean have a selective advantage; extremes are selected against. This reduces variation but maintains the mean (e.g., human birth weight: very large or very small babies have lower survival).
- **Directional selection** — individuals with a phenotype at one extreme of the range have a selective advantage, shifting the mean in that direction (e.g., antibiotic resistance in bacteria, industrial melanism in peppered moths).
- **Disruptive selection** — individuals at both extremes have advantages over those near the mean, potentially leading to a bimodal distribution and eventually speciation (e.g., beak size in some bird populations with two distinct food sources).

Speciation

> **Species:** A group of organisms that can interbreed to produce fertile offspring and are reproductively isolated from other such groups. (Note: this biological species concept has limitations for asexual organisms, fossils, and ring species.)

Allopatric speciation occurs when a population is geographically divided (by a river, mountain range, or other physical barrier). The separated populations experience different selection pressures and genetic drift, accumulating different genetic changes over time. Eventually, the populations become so genetically different that they can no longer interbreed if reunited, forming separate species.

Sympatric speciation occurs within a single population without geographic separation, often through reproductive isolation caused by behavioural differences (mating calls, timing of reproduction), ecological niche separation, or polyploidy (particularly important in plants, where chromosome doubling creates immediate reproductive isolation from the parent population).

Evidence for Evolution

Multiple lines of evidence support evolutionary theory:

- **Fossil record** — shows progressive changes in organisms over geological time, transitional forms (e.g., Archaeopteryx linking dinosaurs and birds), and extinction events
- **Comparative anatomy** — homologous structures (same underlying anatomy, different functions) indicate common ancestry (e.g., pentadactyl limb in mammals, birds, reptiles); analogous structures (different anatomy, similar function) indicate convergent evolution
- **Molecular evidence** — similarities in DNA base sequences, protein amino acid sequences, and genome organisation correlate with evolutionary relatedness; molecular clocks estimate divergence times
- **Biogeography** — distribution of species reflects evolutionary history and geographic barriers (e.g., marsupials concentrated in Australia due to continental drift)
- **Observed evolution** — antibiotic resistance in bacteria, pesticide resistance in insects, and artificial selection in domesticated species demonstrate selection in action

Genetic Drift

Genetic drift is the random change in allele frequency in a population due to chance events rather than natural selection. It has a greater effect in small populations (where chance fluctuations have proportionally larger impacts) and can lead to the fixation (frequency = 100%) or loss (frequency = 0%) of alleles regardless of whether they are beneficial, neutral, or harmful.

Special cases of genetic drift:

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- **Founder effect:** A small group colonises a new area, carrying only a subset of the original population's alleles. The new population may have very different allele frequencies from the parent population. Example: the high frequency of Ellis-van Creveld syndrome among the Amish community of Pennsylvania, founded by a small number of settlers from Germany.
- **Genetic bottleneck:** A catastrophic event (disease, natural disaster, hunting) drastically reduces population size, randomly eliminating many alleles. The surviving population has reduced genetic diversity. Example: northern elephant seals were hunted to fewer than 20 individuals in the 1890s; despite recovery to over 100,000, they show very low genetic variation.

Relationship between drift and selection: In large populations, natural selection is the dominant evolutionary force. In small populations, genetic drift can overpower selection, causing harmful alleles to increase and beneficial alleles to be lost by chance. This is a major concern in conservation biology, as endangered species with small populations are vulnerable to loss of genetic diversity through drift, reducing their ability to adapt to environmental change.

> **Exam Tip:** When comparing natural selection and genetic drift: selection is non-random, directional, and more effective in large populations; drift is random, non-directional, and more significant in small populations. Both change allele frequencies over time but through fundamentally different mechanisms. Selection acts on phenotypes; drift is independent of phenotype.

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